

Date 16/11/18

Page No. 149

Science Notes

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Class 9th

WORK AND ENERGY

✓ Work:- work is done when a force produces motion.

The work done by a force on a body depends on two factors:-

- i) Magnitude of the force
- ii) Distance through which the body moves.

* its S.I. unit is Joule.

* it is a scalar quantity

✓ work done in moving a body is equal to the product of force and the displacement of the body in the direction of force.

So, $\text{Work} = \text{force} \times \text{Displacement in the Direction of force}$

$\text{Work} = \text{force} \times \text{Displacement}$

$$W = F \times S$$

✓ 1 Joule:- When a force of 1 newton moves a body through a distance of 1 metre in its own direction, then the work done is known as 1 Joule.

Date 16/11/18

Page No.: 150

$$1 \text{ Joule} = 1 \text{ newton} \times 1 \text{ metre}$$

$$1 \text{ J} = 1 \text{ Nm}$$

Work is said to be done when:-

- i) a moving object comes to rest.
- ii) an object at rest starts moving
- iii) velocity of an object changes
- iv) shape of an object changes.

NCERT textbook page no. 148 - 149 - Home-work

Q.1

A force of 7N acts on an object. The displacement is say 8m. in the direction of the force. Let us take it that the force acts on the object through the displacement. what is the work done in this case?

Ans

$$F = 7 \text{ N}$$

$$S = 8 \text{ m}$$

$$W = ?$$

$$W = F \times S$$

$$W = 7 \times 8$$

$$W = 56 \text{ Joule Ans}$$

Q.2

when do we say that work is done?

Ans

work is done when a force produces motion in an object. Or work is said to

be done if by applying the force body is displaced

Q.3

Write an expression for the work done

"Facts are facts and will not disappear on account of your likes"

When a force is acting on an object in the direction of its displacement?

Any If a force F acts on an object and S is the displacement of the object in the direction of force, then:-

$$\text{Work done} = \text{Force} \times \text{Displacement}$$

$$W = F \times S$$

Q.4 Define 1 Joule of work.

Any When a force of 1 newton moves a body through a distance of 1 metre in its own direction, then the work done is known as 1 Joule.

$$1J = 1N \times 1m$$

$$1J = 1Nm$$

Q.5 A pair of bullocks exerts a force of 140N on a plough. The field being ploughed is 15m long. How much work is done in ploughing the length of the field?

Any

$$F = 140N$$

$$S = 15m$$

$$W = ?$$

$$W = F \times S$$

$$W = 140 \times 15$$

$$W = 2100 \text{ Joule}$$

Any

✓ Energy:- The energy is the ability to do work.

- * its unit is Joule
- * it is scalar quantity.

$$1 \text{ Kilojoule} = 1000 \text{ Joule}$$

$$1 \text{ kJ} = 1000 \text{ J.}$$

Forms of energy:-

- 1- Kinetic energy
- 2- Potential energy
- 3- chemical energy
- 4- Heat energy
- 5- Light energy
- 6- Sound energy
- 7- Electrical energy
- 8- Nuclear energy

✓ Kinetic energy:- The energy of a body due to its motion is called kinetic energy.
eg:- A moving cricket ball, A moving bullet.

Formula of kinetic energy:-

motion equation of third equation is
$$v^2 - u^2 = 2as$$

$$\text{or } s = \frac{v^2 - u^2}{2a}$$

Date 17, 11, 18

Page No.: 153

We know that work

$$W = F \times S$$

$$[\because F = ma]$$

$$W = m \times \frac{v^2 - u^2}{2g}$$

$$W = m \times \frac{v^2 - u^2}{2}$$

$$W = \frac{1}{2} \times m(v^2 - u^2)$$

$$\because u = 0$$

$$W = \frac{1}{2} m v^2$$

$$\boxed{E_k = \frac{1}{2} m v^2}$$

Potential energy:— The energy of a body due to its position or change in shape is known as potential energy.

Gravitational potential energy:— The energy of a body due to its position above the ground is called gravitational potential energy.

Mechanical energy:— The sum of the potential and kinetic energy is called mechanical energy.

Formula of potential Energy:—

We know that

$$\text{work done} = \text{Force} \times \text{Displacement}$$

$$\because F = mg$$

$$\text{acceleration due to gravity So, } F = mg$$

"You have to take the calculated risk, to earn something."—Dhirubhani Ambani

Date 17, 11, 18

Page No.: 159

$$W = Mg \times h$$

$$W = mgh$$

$$E_p = mgh$$

Mechanical Energy or Constant

$$= mgh + \frac{1}{2} mv^2$$

Law of conservation of energy: —

energy can only be converted from one form to another; it can neither be created nor destroyed.

Home-work

Q-1

Ans

What is the kinetic energy of an object?
The energy of a body due to its motion is called kinetic energy.

Q-2

Ans

Write an expression for the kinetic energy of an object.

Kinetic energy of $= \frac{1}{2} mv^2$ an object.

m = mass of an object.

v = Velocity/speed of the object.

Q-3

The kinetic energy of an object of mass m moving with a velocity of 5 m/s is 25 J .
What will be its kinetic energy when its velocity

is doubled? ~~what~~ what will be its kinetic energy when its velocity is increased three times?

Ans

$$\text{Mass} = ?$$

$$E_k = 25 \text{ J}$$

$$v = 5 \text{ m/s}$$

$$\text{kinetic energy} = \frac{1}{2}mv^2$$

$$25 \text{ J} = \frac{1}{2} \times m \times (5)^2$$

$$25 \times 2 = m \times 25$$

$$\frac{50}{25} = m$$

$$m = 2 \text{ kg}$$

In first case when its velocity is doubled

$$m = 2 \text{ kg}$$

$$v = 5 \times 2 = 10 \text{ m/s}$$

$$E_k = ?$$

$$E_k = \frac{1}{2}mv^2$$

$$E_k = \frac{1}{2} \times 2 \times (10)^2$$

$$E_k = 100 \text{ J} \quad \text{Ans}$$

in second case, when its velocity is tripled

$$m = 2 \text{ kg}$$

$$v = 5 \times 3 = 15 \text{ m/s}$$

$$E_k = \frac{1}{2}mv^2$$

$$E_k = \frac{1}{2} \times 2 \times (15)^2$$

$$E_k = 225 \text{ J} \quad \text{Ans}$$

Date 19/11/18

Page No.: 156

Power: — Power is defined as the rate of doing work.

*
$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}} \quad \text{or} \quad P = \frac{W}{t}$$

* Power is a scalar quantity.
* its S.I. unit is watt.

average power: — power is also defined as the rate at which energy consumed.

$$P_{av} = \frac{\text{Energy consumed}}{\text{Time taken}}$$

$$P_{av} = \frac{E}{t}$$

Units of power: —

1 watt is the power of an appliance which consumes energy at the rate of 1 Joule per Second.

$$1 \text{ watt} = \frac{1 \text{ Joule}}{1 \text{ second}}$$

$$\text{or } 1 \text{ W} = \frac{1 \text{ J}}{1 \text{ s}}$$

$$1 \text{ watt} = 1 \text{ J/s}$$

$$1 \text{ kW} = 1000 \text{ W}$$

$$1 \text{ Mega watt} = 1000000 \text{ W}$$

$$1 \text{ m W} = 10^6 \text{ W}$$

$$1 \text{ horse power} = 746 \text{ watt}$$

"Facts are facts and will not disappear on account of your likes." — Jawaharlal Nehru

Date 19/11/18

Page No.: 157

✓ Commercial unit of energy: — The Commercial unit of energy is kilowatt.
* its S.I. unit of Joule.

Question: What is 1 kW/h .

Ans: The energy used in one hour at the rate of 1 kW is called 1 kW/h .

Relation between kilowatt-hour and joule:

1 kilowatt-hour is the amount of energy consumed at the rate of 1 kilowatt for 1 hour .

$1 \text{ kilowatt-hour} = 1 \text{ kilowatt for } 1 \text{ hour}$

or $1 \text{ kilowatt hour} = 1000 \text{ watts for } 1 \text{ hour}$

— (I)

But:- $1 \text{ watt} = \frac{1 \text{ Joule}}{1 \text{ Second}}$

For equation Ist

$1 \text{ kilowatt hour} = 1000 \times \frac{\text{Joules}}{\text{Seconds}} \text{ for } 1 \text{ hour}$

$\therefore 1 \text{ Hour} = 60 \times 60 \text{ Seconds}$

$\therefore 1 \text{ Kw/h} = 1000 \times \frac{\text{Joules}}{\text{Seconds}} \times 60 \times 60 \text{ Seconds}$

$1 \text{ Kw/h} = 36,00,000 \text{ J.}$

or $1 \text{ Kw/h} = 3.6 \times 10^6 \text{ J}$

* EXERCISE *

Q.1

Look at the

Q.3

A battery lights a bulb. Describe the energy changes involved in the process.

Ans

Chemical energy $\rightarrow$ Electrical energy $\rightarrow$
Heat energy $\rightarrow$ Light energy.

Q.4

Certain

force.

 $\rightarrow$

in first case:—

$$M = 20 \text{ kg}$$

$$V = 5 \text{ m/s}$$

$$K.E. = \frac{1}{2} m v^2$$

$$K.E. = \frac{1}{2} \times 20 \times (5)^2$$

$$K.E. = 10 \times 25$$

$$K.E. = 250 \text{ J}$$

in second case

$$M = 20 \text{ kg}$$

$$V = 2 \text{ m/s}$$

$$K.E. = \frac{1}{2} \times m v^2$$

$$K.E. = \frac{1}{2} \times 20 \times (2)^2$$

$$E_K = 10 \times 4$$

$$E_K = 40 \text{ J}$$

work done by force = change in kinetic energy

$$= 250 - 40$$

$$= 210 \text{ J}$$

"Facts are facts and will not disappear on account of your likes." — Jawahar Lal Nehru

Q-5 An object - - - - - Your answer.

Ans The work done is zero, This is because the gravitational force and displacement are perpendicular to each other.

Q-6 The potential - - - - - ? Why?

Ans It does not violate the law of Conservation of energy. whatever, is the decrease in potential energy due to loss of height, same is the increase in the K.E. due to increase in velocity of the body.

Q-7 What are the various energy transformation that occur when you are riding a bicycle?

Ans The chemical energy of the food changes into heat and then to muscular energy on peddling, The muscular energy changes into mechanical energy.

Q-8 Does the transfer - - - - - going?

Ans Energy transfer does not take place as no displacement takes place in the direction of applied force. The energy spent is used to overcome inertia of rest of the rock.

Q.9 A certain household has consumed 250 units of energy during a month. How much energy is this in joules?

Ans Energy consumed = 250 units
= 250 Kw/h

$$1 \text{ Killowatt hour} = 3.6 \times 10^6$$

$$250 \text{ Killowatt hour} = 3.6 \times 10^6 \times 250$$
$$= 9 \times 10^8 \text{ Joules}$$

Ans

Q.11 What is the work done by the force of gravity on a satellite moving round the earth? Justify your answer.

Ans When a satellite moves round the earth, then at each point of its path, the direction of force of gravity on the satellite is perpendicular to the direction of its displacement. Hence, the work done on the satellite by the force of gravity is zero.

Q.13 A person holds a bundle of hay over his head for 30 minutes and gets tired. Has he done some work or not? Justify your answer.

Ans The person does not do work because no displacement takes place in the direction of applied force as the force acts in the vertically upward direction.

Date 21/11/18

Page No.: 161

Q. 14 An electric heater is rated 1500 W. How much energy does it use in 10 hours?

Ans $P = 1500 \text{ W} = \frac{1500}{1000} = 1.5 \text{ kW}$

$t = 10 \text{ h}$

Power = $\frac{\text{Energy Consumed}}{\text{time}}$

$1.5 = \frac{\text{Energy Consumed}}{10}$

Energy Consumed = 1.5×10
= 15 units Ans

Q. 16 An object - - - - - to rest?

Ans Kinetic energy of an object of mass 'm' moving with a velocity 'v' is given by the expression $\frac{1}{2}mv^2$. To bring the object to rest, an equal amount of work i.e. $\frac{1}{2}mv^2$ is required to be done on the object.

Q. 17 Calculate - - - - - 60 km/h?

Ans

$M = 1500 \text{ kg}$

$V = 60 \text{ km/h} = 60 \times \frac{5}{18} = \frac{50}{3} \text{ m/s}$

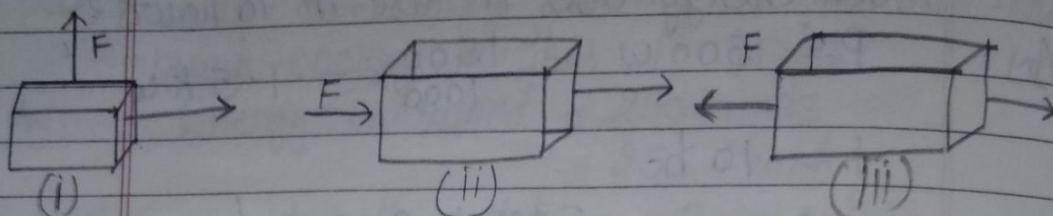
$K_E = \frac{1}{2}mv^2$

$K_E = \frac{1}{2} \times 1500 \times \left(\frac{50}{3}\right)^2$

$K_E = \frac{750 \times 2500}{9} = 208333.3 \text{ J}$ Ans

"You have to take the calculated risk, to earn something." - Dhirubhai Ambani

Q.18 In each of the - - - - - Zero.



- i) The direction of force F and the direction of displacement are at right angles to each other. So, the work done is zero.
- ii) the displacement is in the direction of force F . So the work done is positive.
- iii) the force F acts in a direction opposite to the direction of displacement, So the work done is negative.

Q.19 Soni says - - - - - ? why?

Ans Yes, acceleration in an object could be zero even when several forces are acting on it. This happens when all the forces cancel out each other i.e. the net force acting on the object is zero.

Q.20 Find the energy in kWh consumed in 10h by four devices of power 500w each.

Ans

Power of one device = 500w

Power of four device = 500×4
= 2000

= $\frac{2000 \times 10}{1000} = 20 \text{ kWh}$

Date 21/11/18

Page No.: 163

$$T = 10h$$

$$P = \frac{\text{Energy Consumed}}{\text{Time}}$$

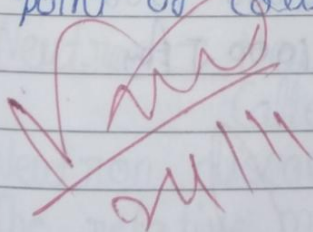
$$2 = \frac{\text{Energy Consumed}}{10}$$

$$\text{Energy Consumed} = 10 \times 2 = 20 \text{ units}$$

Q.21) A freely falling object eventually stops on reaching the ground. what happens to its kinetic energy.

Ans As the object hits the hard ground its kinetic energy gets converted into:-

- i) heat energy (the object and the ground become slightly warm)
- ii) Sound energy (sound is heard when the object hits the ground)
- iii) Potential energy of compression of the body and the ground (the object and the ground get deformed a little bit at the point of collision).



class-

$\Rightarrow m = 40 \text{ kg} \quad g = 10 \text{ m/s}^2 \quad h = 5 \text{ m}$

$$E_p = M \times g \times h$$

$$= 40 \times 10 \times 5 = 2000 \text{ Joules Ans}$$

the object is at a height of 5m. when this object is allowed to fall and it is half way down.

$$h = \frac{5}{2} = 2.5$$

$$M = 40 \text{ kg}$$

$$E_p = m \times g \times h$$

$$= 40 \times 10 \times 2.5 = 1000 \text{ Joules}$$

Law of Conservation of energy:-

Total potential energy = potential energy + kinetic energy

2000 - 1000 = Kinetic energy

$$\text{Kinetic Energy} = 2000 - 1000 = 1000 \text{ J}$$